

Projected Krylov Sign Descent and the Structure of Ground-State Sign Basins

A compact mechanism note for the square-lattice J1-J2 and triangular Heisenberg tests. [Chatty subproject](#) | current repository head at preparation: 328bc37

Abstract

A one-step Krylov/local-energy rule was observed to reconstruct the frustrated J1-J2 ground-state signs with unexpectedly high accuracy. Exact finite-cluster tests show that this is the first strong contraction step of a nonlinear projected-Krylov sign descent. At fixed exact amplitudes, repeated label-free threshold updates reach the exact square-lattice signs even from poor Marshall starts. The ground-state attraction basin is empirically large under random sign corruption. The triangular model does not invalidate local contraction: instead, structured gauges can occupy competing basins. One such obstruction is exact and analytic - the update preserves symmetry character whenever the amplitude modulus is symmetry invariant.

1. The object being iterated

Write the wavefunction as a positive modulus times a binary sign field, $\psi_s(x) = a(x)s(x)$, with $a(x) \geq 0$ and $s(x) = \pm 1$. For the mechanism tests, $a(x)$ is held fixed to the exact ground-state modulus.

$$r_s(x) = [H \psi_s](x) / \psi_s(x)$$

A single Krylov vector obeys $[(t-H)\psi_s](x) = [t - r_s(x)]\psi_s(x)$. Therefore its sign is exactly the threshold rule

$$T_t[s](x) = s(x) \operatorname{sign}[t - r_s(x)].$$



At every iteration we choose t without using hidden ground-state signs: among all realizable thresholds of r_s , we select the one with the lowest fixed-amplitude variational energy and then project back to the same modulus $a(x)$.

Key exact fact: monotone projected descent

The threshold family always contains the current sign pattern: choosing t above $\max_x r_s(x)$ leaves every sign unchanged. Hence the energy-minimizing update satisfies $E[T(s)] \leq E[s]$. With the exact ground-state modulus, the exact signs are a global minimum and a fixed point because $r_*(x) = E_0$ for every nonzero component.

2. What the square-lattice tests establish

On the periodic 4x4 square-lattice spin-1/2 J1-J2 model in $S^z=0$, exact diagonalization provides both the modulus and reference signs. The first step is spectacular near $J2/J1=1/2$, but the stronger result is that repeated updates converge all the way to the exact signs.

J2/J1	start	sign-overlap trajectory O_S	exact at
0.5	Marshall	0.97454 -> 0.999319 -> 0.999998 -> 1	3 updates
0.6	Marshall	0.79557 -> 0.95713 -> 0.99127 -> 0.99944 -> 1	4
0.8	Marshall	0.12366 -> 0.06197 -> ... -> 0.96288 -> 0.99120 -> 0.99985 -> 1	10
1.0	Marshall	0.06421 -> 0.17135 -> 0.29020 -> 0.33926 -> 0.91039 -> ... -> 1	9
1.0	stripe	0.95582 -> 0.99488 -> 0.999692 -> 0.9999994 -> 1	4

Interpretation. A good gauge is not required for eventual convergence on this tested square cluster. It mainly makes the first step exceptionally efficient. Even Marshall at $J2/J1=0.8$ and 1.0, where the first step is poor, eventually reaches the exact signs.

Why one step looked miraculous. At $J2/J1=0.5$ the Marshall state is already positioned so that r_0 orders nearly all physically important residual sign defects. The first threshold removes almost all of them at once; recomputing r then exposes smaller residual defects. Thus $0.97454 \rightarrow 0.999319 \rightarrow 0.999998 \rightarrow 1$ is rapid basin contraction, not a one-off accident.

3. The attraction basin is large, but distance is not the whole story

To quantify 'start close', the exact sign pattern was randomly corrupted until a chosen fraction q of the ground-state probability weight carried the wrong sign. The same label-free iteration was then applied. Across $J_2/J_1=0.5, 0.6$, and 1.0 , every one of 32 random starts recovered for $q \leq 0.42$. Since O_{initial} is approximately $|1-2q|$, this includes starts with sign overlap only about 0.16.

Above this range the outcome becomes strongly pattern- and model-dependent. The correct statement is therefore not a rigorous spherical 'basin radius'. Rather, the ground-state basin occupies a large region under random physical-weight sign corruption, while structured sign patterns may sit in qualitatively different basins.

4. Triangular lattice: competing basins, not failure of contraction

For the nearest-neighbor triangular Heisenberg antiferromagnet on a periodic 6×3 torus (18 spins, $S^z=0$), three simple two-color gauges do not approach the exact sign structure under repeated updates. They flow to wrong threshold-stable fixed points with essentially zero physical sign overlap.

initial structured gauge	terminal behavior	ground-state sign overlap
maxcut_x	wrong fixed point after 13 updates	$\sim 2 \times 10^{-16}$
parity_y	wrong fixed point after 18 updates	$\sim 1 \times 10^{-15}$
parity_xy	wrong fixed point after 18 updates	$\sim 2 \times 10^{-15}$

However, this is not because the triangular ground-state signs are locally unstable under the map. Starting from randomly corrupted versions of the exact signs, all 8/8 trials recovered through $q=0.45$, and 4/8 recovered even at $q=0.49$. Thus the exact sign state is also a strong attractor on this cluster; the structured gauges simply begin in other basins.

Exact basin invariant: symmetry character

Let P be a symmetry with $[P,H]=0$. If the fixed modulus is invariant, $P a=a$, and the current fixed-amplitude wavefunction has character $P \psi_s = \chi \psi_s$, then $r_s(Px)=r_s(x)$. Any threshold mask $\text{sign}(t-r_s)$ is therefore P -invariant, so every projected-Krylov update preserves χ . **A start in the wrong exact symmetry sector can never reach the target sector.**

The numerical translation audit makes this concrete. On the triangular cluster the exact ground state has $T_x=+1$ and $T_y=+1$. The maxcut_x and parity_xy gauges have $T_x=-1$ (residuals of order 10^{-12}), so they are rigorously excluded from the ground-state basin. By contrast, on the square 4×4 cluster the exact state, Marshall gauge, and stripe gauge all have $T_x=T_y=+1$, consistent with eventual recovery even when their initial overlap is poor.

The parity_y triangular fixed point is not explained by this simple translation-character theorem. It is the clean remaining example of a competing basin inside, or across a mixture of, compatible symmetry content.

5. What is established, and what is not

Established on the tested exact clusters

(i) The update is exactly the sign of a Krylov vector followed by projection back to fixed amplitudes. **(ii)** Energy-minimizing threshold selection is monotone in the fixed-amplitude energy. **(iii)** The square-lattice ground-state signs are reached exactly from every tested physical start, including poor Marshall starts at large J_2 . **(iv)** The ground-state basin is empirically large under random corruption. **(v)** Exact symmetry characters are invariant and can split sign space into disconnected basins.

Still open

This is not yet a scalable ground-state algorithm. The controlled tests use the exact ground-state modulus and evaluate the threshold objective over the finite Hilbert space. The immediate theory problem is to explain wrong fixed points not already symmetry-forbidden (notably triangular parity_y), likely through weighted sign flux/frustration or another invariant. The immediate algorithmic problem is to determine how much of the contraction survives with sampled/approximate amplitudes and stochastic threshold-energy estimates.

6. Reproducibility and live project links

[Basin dynamics verdict](#) | [Symmetry-sector invariant](#) | [Current STATUS](#)

Primary scripts: [iterated square test](#), [wrong-gauge iteration](#), [basin scan](#), [triangular iteration](#), [symmetry audit](#).

Internal research snapshot; finite-size and exact-amplitude assumptions are explicit.